

## 5-1 Cleaning the Cooling Fan in the Motor Rear Hood



- ☑ To prevent fire, explosion or accidents harmful to safety, the surrounding air should be kept free at all times when using solvents.
- ☑ In narrow working places (such as coal pits), artificial respiration devices with centralized air supply and masks with hoses or self-contained breathing equipment must be provided for each user.
- ☑ Users must wear goggles, aprons and gloves. Solvents and their volatile gases must not be exposed to burning flames or sparks and should be stored in safe and qualified containers.

During the operation, the environment should be kept clean, dry, and well-ventilated. Dust and cotton on the surface of the motor should be cleaned regularly (every 10 days) to keep the surface of the motor clean. Clean the oil dirt and foreign objects on the bottom of the air hood and the cooling fan to avoid affecting the cooling effect and reducing the product lifespan of the cooling fan.

(A) When a motor is used for a long time, there must be dust, carbon powder, grease, etc. adhered to the motor. These foreign objects can easily absorb the moisture and cause insulation degradation. The cooling effect will be hindered and causes the temperature to rise, thus causing accidents. Therefore, it is suggested to perform regular inspections.

(B) Cleaning methods and precautions:

- ① Use air compressor and small air blower:
  - a. Make sure that there's no moisture in the compressed air.
  - b. The pressure should not exceed  $4\text{kg/cm}^2$  to avoid damaging the insulation of the coil. However, if the pressure is too low, the effect will be poor. The most appropriate pressure is about  $4\text{kg/cm}^2$ .
- ② Vacuum cleaning:
 

Vacuum cleaning can be done before or after other cleaning methods to remove dust or foreign objects. This is an effective way to remove dirt from the winding surface without causing dust to fly everywhere. The tools used for vacuum cleaning must be non-metallic to prevent damage to the winding insulation.
- ③ Wiping:
 

The dirt on the winding surface can be wiped off with soft, lint-free clothes. If the dirt contains oil, the wiping clothes can be dipped in some safe petroleum solvent (not so wet that it will drip out). Use inhibitory chemicals in hazardous areas. When using inhibited methyl chloroform as a solvent, you must be careful and remove it from the operation site immediately after use. Although this solvent will not burn under normal circumstances, it is still necessary to take appropriate health and safety measures due to its toxicity.